Lecture 03 - Introduction -READ FIRST-

Lecture 03 - Describing and Summarizing Data - Introduction

Purpose of the Lesson

This lesson introduces descriptive statistics: the tools we use to turn raw data into information that can be summarized, visualized, and interpreted carefully.

In finance and business, data usually arrive as observations: prices, returns, transactions, customers, invoices, ratings, sectors, or portfolio values. These observations are useful, but they rarely speak for themselves. Before we can estimate, test, forecast, or model anything, we first need to understand what the dataset contains, what each observation represents, what each variable measures, and which summaries are meaningful.

The lecture also introduces one of the most important transformations in empirical finance: moving from prices to returns. Prices tell us the value of an asset at a point in time. Returns tell us the gain or loss relative to the amount invested. For most statistical work in finance, returns are the more useful object because they allow us to compare assets, summarize performance, and measure risk.

Python is used only as a practical support tool. The goal is not to turn this lecture into a programming lesson. The goal is to understand the structure of data and to use simple code when it helps us inspect, transform, summarize, and visualize that data.

Learning Objectives

This lecture aims to:

1. Introduce the statistical language used to describe datasets: population, sample, observation, variable, statistic, parameter, estimator, and estimate.
2. Explain why the type of variable matters for choosing appropriate summaries and graphs.
3. Show how raw observations can be transformed into frequency tables, bar charts, and histograms.
4. Introduce the main measures of location and dispersion: mean, median, mode, percentiles, quartiles, range, variance, standard deviation, and coefficient of variation.
5. Develop the price-to-return logic used throughout empirical finance.
6. Explain how average returns and volatility can be annualized, and what assumptions are involved.

Learning Outcomes

By the end of this session, students should be able to:

1. Describe a dataset using precise statistical language.
2. Distinguish a population parameter from a sample statistic, an estimator, and an estimate.
3. Classify variables as categorical, ordinal, binary, discrete numerical, or continuous numerical.

4. Choose appropriate descriptive summaries and visualizations for different types of variables.
5. Compute and interpret measures of location and dispersion.
6. Compute simple returns and log returns from price data.
7. Explain why simple returns compound and log returns add.
8. Annualize average returns and volatility using the appropriate rules.
9. Report descriptive statistics carefully, making clear that they summarize a historical sample rather than guarantee future outcomes.

Lesson Roadmap

The lecture begins with the basic structure of a dataset. We distinguish the population or process we care about from the sample we actually observe. We then define observations and variables, and we explain why rows and columns in a data table have statistical meaning.

The next step is statistical vocabulary. You will learn the difference between a parameter, a statistic, an estimator, and an estimate. This distinction is essential because most statistical work in finance uses observed sample data to say something cautious about a broader and uncertain process.

The lecture then turns to types of data. A variable can be nominal categorical, ordinal categorical, binary, discrete numerical, or continuous numerical. This classification matters because it determines which summaries and graphs make sense. A sector label, a satisfaction rating, a default indicator, a number of purchases, and a daily return should not all be treated in the same way.

After that, we move from raw data to descriptive summaries. Categorical and discrete variables can often be summarized with frequency tables and bar charts. Continuous numerical variables are usually grouped into intervals and visualized with histograms. These tools help us see the shape of the data before reducing it to a few numbers.

The second part of the lecture focuses on location and dispersion. Measures of location, such as the mean, median, and mode, describe where the data are centered. Measures of dispersion, such as the range, variance, standard deviation, interquartile range, and coefficient of variation, describe how spread out the observations are. In finance, dispersion is especially important because it is closely related to risk.

The final part applies these ideas to financial prices and returns. You will learn how to compute simple returns and log returns, why a return series has one fewer observation than the original price series, how simple returns compound over time, and why log returns add. The lecture ends by showing how average returns and volatility are annualized.

Central Idea

The central idea of the lecture is:

Descriptive statistics are not just calculations. They are disciplined summaries of a specific sample, and they must be interpreted in context.

For example, suppose the average daily return of a stock in a historical sample is 0.11%. A careless statement would be:

The stock earns 0.11% per day.

A more careful statement is:

The average daily return in the observed sample is 0.11%. This is a sample statistic computed from historical data, not a guarantee of future performance.

This distinction will appear repeatedly in the course. It is the foundation for sampling variability, confidence intervals, hypothesis testing, regression inference, and risk measurement.

How to Work Through the Lesson

Begin with the lecture notes and slides. Focus first on the vocabulary: population, sample, observation, variable, parameter, statistic, estimator, and estimate. These terms will be used throughout the rest of the course.

Then pay attention to variable types. Before calculating a statistic, ask what kind of variable you are working with. A mean is meaningful for invoice amounts or returns, but not for industry names. A frequency table is natural for categories, while a histogram is natural for continuous numerical data.

Next, work through the examples on location and dispersion. Do not only memorize formulas. Make sure you can explain what each statistic tells you. The mean describes a typical level, but it can be sensitive to outliers. The median is based on position. The standard deviation describes typical dispersion around the mean. Percentiles describe positions in the ordered data.

Finally, work through the price-to-return workflow. Make sure you understand why returns are computed from pairs of prices, why the first return is missing, why adjusted prices are usually used, and why returns are more comparable than raw price changes.

After the lecture, complete the exercises. Some exercises are conceptual, while others use Python. When checking your work, pay attention not only to numerical answers but also to interpretation.

What to Pay Attention To

Three habits matter in this lecture.

First, always identify the statistical object. Ask: What is the sample? What is one observation? What are the variables? What broader quantity might we want to estimate?

Second, match the summary to the variable. A bar chart is appropriate for categories. A histogram is appropriate for a numerical distribution. A mean is appropriate for numerical data, but not for arbitrary labels. A standard deviation is meaningful only when numerical distances are meaningful.

Third, use careful language. Descriptive statistics summarize the sample. They do not prove what will happen in the future. A sample average return is an estimate, not a promise. A sample standard deviation is a historical measure of dispersion, not a complete description of future risk.

End-of-Lesson Check

By the end of the lecture, you should be comfortable answering questions such as:

1. What is the difference between a parameter and an estimate?
2. What is the difference between a statistical variable type and a Python storage type?
3. Why is the average of invoice IDs meaningless, even if invoice IDs are stored as numbers?
4. When should we use a bar chart, and when should we use a histogram?

5. Why can two datasets have the same mean but very different risk?
6. What is the difference between variance and standard deviation?
7. Why can numpy.std() and pandas .std() produce slightly different results by default?
8. Why does a return series have one fewer observation than the original price series?
9. Why are returns usually more useful than raw price changes?
10. Why do simple returns compound while log returns add?
11. Why is annual volatility multiplied by the square root of 252 rather than by 252?
12. How should we interpret a historical sample average return?

If these questions are clear, you are ready to move into the probability and random-variable concepts that come next.